

Association of obesity and nutritional status on early sepsis-associated acute kidney injury in older patients with sepsis

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TITLE PAGE

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Association of obesity and nutritional status on early sepsis-associated acute kidney injury in older patients with sepsis

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Abstract

Background: Despite numerous studies have examined the impact of obesity and poor nutritional status on the prognosis of critically ill patients, their relationship with sepsis-associated acute kidney injury (SA-AKI), particularly in older patients,

remains unclear. This study aimed to investigate the association between obesity, nutritional status, and early SA-AKI in older patients with sepsis.

Methods: This retrospective cohort study utilized data from the Medical Information Mart for Intensive Care IV (MIMIC-IV) database. Older septic patients without pre-existing chronic kidney disease (CKD) stage 3–5 or elevated serum creatinine (SCr; >1.2 mg/dL in males and >1.1 mg/dL in females) at admission were included to focus on incident early SA-AKI. Obesity was measured by body mass index (BMI), and nutritional status was evaluated using the Geriatric Nutritional Risk Index (GNRI) and Prognostic Nutritional Index (PNI) as prognostic indicators. Multivariable logistic regression, multivariable fractional polynomial regression, and restricted cubic spline models were applied to analyze the associations between BMI, GNRI, PNI, and early SA-AKI. Additionally, the relationships between BMI, GNRI, PNI and outcomes were examined in early SA-AKI patients.

Results: Among 4238 older septic ICU patients without pre-existing renal dysfunction, approximately 50% developed SA-AKI within 48 hours of Intensive care unit (ICU) admission. Overweight (adjusted odds ratio [*AOR*] 1.39, 95% confidence interval [*CI*] 1.17–1.65), obese (*AOR* 1.95, 95% *CI* 1.63–2.35), and severely obese (*AOR* 2.19, 95% *CI* 1.63–2.94) were associated with higher odds of early SA-AKI risk compared with normal weight patients, while each 1-point increase in BMI raised risk of early SA-AKI by 4% (95% *CI* 1.03–1.05). No significant correlations were found between GNRI or PNI and early SA-AKI risk. Among older patients with early SA-AKI, underweight (in-hospital mortality: *AOR* 2.06, 95% *CI* 1.07–3.96; 6-month mortality: adjusted hazard ratio [*AHR*] 1.88, 95% *CI* 1.21–2.92) and lower GNRI (in-hospital mortality: *AOR* 1.80, 95% *CI* 1.00–3.22; 6-month mortality: *AHR* 1.38, 95% *CI* 1.06–1.81) predicted higher in-hospital and 6-month mortality.

Conclusion: In a selected cohort of older septic ICU patients without pre-existing renal dysfunction, overweight and obesity are associated with a higher risk of early SA-AKI. Once SA-AKI develops, underweight and poorer nutritional status are related to worse survival. These findings highlight the prognostic value of BMI and nutritional indices for risk stratification in this population. Further prospective studies are warranted to validate these associations and explore their potential implications for risk stratification.

Keywords: Acute kidney injury; Older patients; Obesity; Geriatric Nutritional Risk Index; Prognostic Nutritional Index; Critical care; Sepsis

Background

Sepsis, a life-threatening condition characterized by organ dysfunction due to a dysregulated host immune response to infection, remains one of the leading causes of mortality in Intensive care units (ICUs) worldwide [1]. The kidneys are among the first organs to be affected in sepsis, making acute kidney injury (AKI) one of the most common complications. In patients with septic shock, the incidence of sepsis-associated acute kidney injury (SA-AKI) can reach up to 60% [2]. SA-AKI significantly increases the risk of developing chronic kidney disease (CKD) and the need for renal replacement therapy (RRT), while also elevating both short-term and long-term mortality rates [3, 4].

Older patients constitute an increasing portion of patients with SA-AKI, often facing poorer prognoses and higher mortality rates compared to younger individuals [5, 6]. The natural decline in kidney function, a diminished immune response, and the presence of multiple comorbidities make older adults particularly vulnerable to SA-AKI, necessitating close monitoring and timely intervention. Early identification of high-risk older patients is therefore critical for clinicians to make accurate assessments and implement appropriate treatment strategies.

Obesity and nutritional status are key factors in older patients with sepsis, as they significantly influence the development and progression of AKI. According to the Intensive Care Over Nations audit, approximately 20% of ICU patients are obese [7, 8]. Obesity is commonly associated with a chronic inflammatory state, which can exacerbate organ damage in sepsis [9]. Obesity is also related to metabolic abnormalities and hemodynamic changes that may increase the risk of AKI [10]. However, numerous real-world studies have reported the “obesity paradox”, a phenomenon where obese patients sometimes exhibit improved survival rates and more favorable outcomes, despite obesity being traditionally considered as a risk factor for poor outcomes [8, 11]. One explanation for this paradox is that obese patients may have greater nutritional reserves, which help them better withstand the catabolic stress of critical illness [11]. The survival advantage of obesity has been observed in patients with cardiovascular disease, heart failure, stroke, diabetes, chronic obstructive pulmonary disease (COPD), and cancer [12–15]. However, it remains unclear whether the “obesity paradox” applies to the risk of early SA-AKI, particularly in older patients who are more likely to have sarcopenic obesity [16, 17].

Poor nutritional reserve is common in critical ill older patients and is associated with adverse outcomes such as increased mortality, prolonged mechanical ventilation, and higher infection rates [18]. A meta-analysis of 1168 patients across 20 studies revealed that 38% to 78% of ICU patients suffer from poor nutritional status [18]. Poor nutritional status can coexist with obesity, particular in older adults, due to inadequate intake or absorption of protein and calories, a condition known as protein-energy malnutrition. The World Health Organization (WHO) has highlighted the “double burden of malnutrition”, which refers to the simultaneous presence of malnutrition and either obesity or diet-related diseases, as a pressing global health challenge [19]. Understanding the balance between obesity and malnutrition is crucial, as both conditions may coexist in older patients, and their combined effects on the risk of AKI are not yet fully understood.

Geriatric Nutritional Risk Index (GNRI) and Prognostic Nutritional Index (PNI) are important nutrition-related prognostic indices for risk stratification, especially when formal diagnoses of malnutrition are unavailable. GNRI, which is derived from albumin levels and weight relative to ideal weight, is a marker of chronic nutrition-related risk in older adults. PNI, a composite index based on serum albumin and lymphocyte count, is associated with protein-energy reserve and immune-inflammatory status [20]. A retrospective study of 12,058 patients found that a low GNRI was significantly associated with increased in-hospital mortality, prolonged ICU stay, and longer overall hospitalization in critically ill patients with AKI [21]. Moreover, studies have shown that a higher PNI score correlates with a reduced risk of AKI, postoperative infection, and mortality [22, 23].

Nevertheless, whether the BMI-defined obesity and these nutrition-related prognostic indices are associated with the risk of early SA-AKI and subsequent outcomes in older septic patients remains insufficiently studied. This study aims to examine the associations of BMI, GNRI, and PNI with early SA-AKI and subsequent mortality in older septic ICU patients, with the goal of providing a multidimensional and clinically applicable risk stratification framework that helps identify older patients at high risk for early SA-AKI.

Methods

Study design and patients

This study employed a retrospective cohort design using data from the Medical Information Mart for Intensive Care IV (MIMIC-IV, v2.2) database, which includes

information on over 70,000 patients admitted to the ICUs at the Beth Israel Deaconess Medical Center in Boston, Massachusetts, between 2008 and 2019 [24]. In this analysis, we selected data from 4238 older patients who were diagnosed with early sepsis upon ICU admission. The specific inclusion criteria were: age ≥ 65 years, ICU stay of at least 48 hours, and a sepsis diagnosis according to the Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3) [1] within 48 hours of ICU admission. For patients with multiple ICU admissions, only the first admission record from their initial hospitalization was included in the analysis. Patients with pre-existing CKD stage 3–5 documented in the MIMIC-IV database were excluded to focus on incident early SA-AKI. Additionally, patients with renal dysfunction at the time of ICU admission, defined as a baseline serum creatinine (SCr) level greater than 1.2 mg/dL for males and greater than 1.1 mg/dL for females, were also excluded. This exclusion criterion aimed to reduce the risk of misclassifying pre-existing kidney dysfunction as AKI, in line with prior studies focusing on incident early SA-AKI [25, 26]. Patients with missing data on weight and height values were excluded from the analysis. The flowchart outlining the patient inclusion and exclusion is presented in Fig 1.

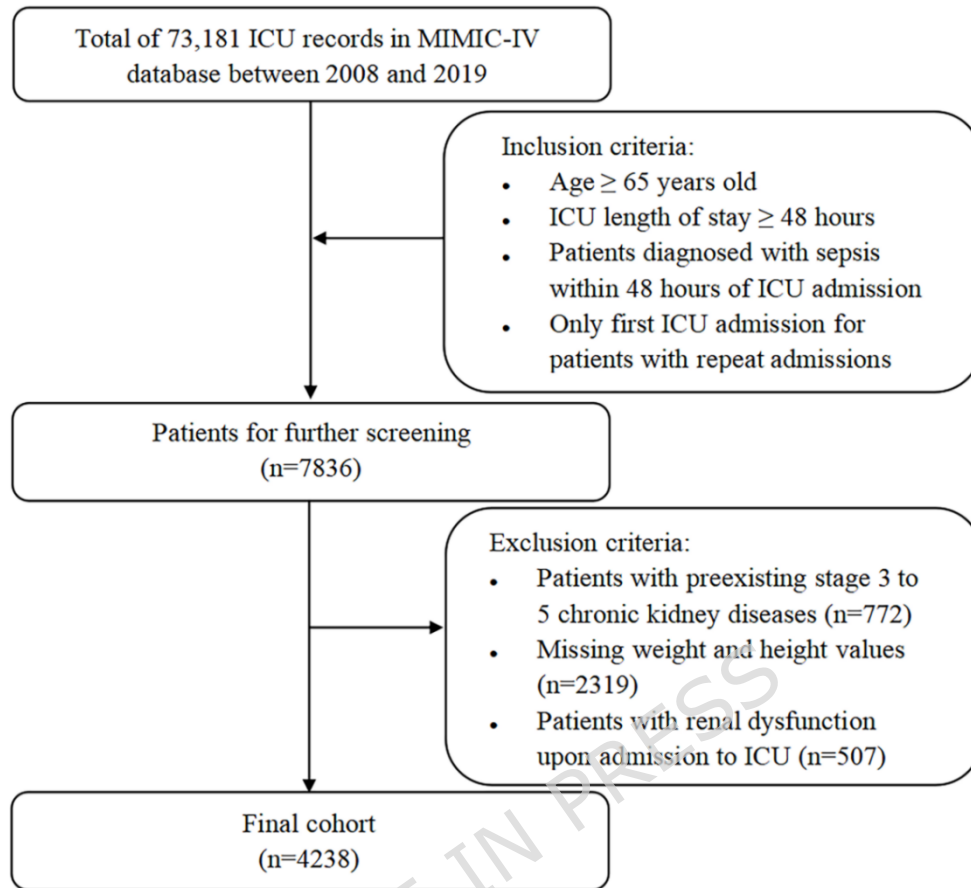


Fig 1 Patient enrollment flowchart

Variables and outcomes

The primary outcome of this study was early SA-AKI, defined as AKI occurring within 48 hours of ICU admission in patients with sepsis diagnosed within the same 48-hour window [27]. Sepsis was diagnosed according to the Sepsis-3 criteria [1], while AKI was diagnosed based on the Kidney Disease Improving Global Outcomes (KDIGO) criteria. Specifically, AKI was defined as any of the following: an increase in SCr by ≥ 0.3 mg/dL within 48 hours, a ≥ 1.5 -fold increase in SCr from baseline within the previous 7 days, or a urine output of < 0.5 mL/kg/h for 6 hours or longer [28]. The increase was calculated using the maximum SCr within the 48-hour window relative to baseline (defined as the SCr at ICU admission). Due to the incomplete and intermittent documentation of urine output in the MIMIC-IV database, which often lacks the reliable hourly measurements necessary to operationalize the KDIGO urine output threshold, AKI in this study was identified solely based on changes in SCr levels to ensure consistent and reproducible case ascertainment across patients.

This approach is consistent with previous ICU database studies [29, 30]. AKI was further categorized into three stages according to the KDIGO criteria: Stage 1 involved an absolute rise in SCr of ≥ 0.3 mg/dL, or a >1.5 - to 2-fold increase from baseline; Stage 2 was characterized by a >2 - to 3-fold increase from baseline; and Stage 3 was defined by a SCr level of ≥ 4.0 mg/dL, a >3 -fold increase from baseline, or the requirement for kidney replacement therapy [28]. Secondary outcomes included in-hospital mortality, length of ICU stay, length of hospital stay, and survival up to 6 months after ICU admission among patients with early SA-AKI.

BMI, calculated as weight (kg) divided by height (m)², was categorized according to the guidelines of the Centers for Disease Control and Prevention in the United States (U.S.) as underweight (<18.5), normal weight (18.5 to 24.9), overweight (25.0 to 29.9), obesity (30 to 39.9), and severe obesity (≥ 40.0) [31].

The GNRI and PNI were used as surrogate prognostic indicators associated with nutrition-inflammation status and physiologic reserve, which are particularly relevant in older patients with sepsis who experience catabolic stress and immune dysregulation [26, 32]. GNRI was calculated using the formulas (1) to (3). Based on previous studies, GNRI was categorized into four risk groups: major risk (GNRI <82), moderate risk ($82 \leq \text{GNRI} < 92$), low risk ($92 \leq \text{GNRI} \leq 98$), and no risk (GNRI >98) [33]. PNI was calculated using the formula (4) and categorized into four levels based on its quartiles. The data used to compute GNRI and PNI, including albumin levels, height, weight, and total lymphocyte count, were collected from baseline measurements taken upon ICU admission, typically within the first several hours.

$$\text{GNRI} = 1.489 \times \text{Albumin} \left(\frac{\text{g}}{\text{L}} \right) + 41.7 \times \frac{\text{Current Weight (kg)}}{\text{Ideal Weight (kg)}} \quad (1)$$

$$\text{Ideal Weight for men} = \text{Height(cm)} - 100 - \frac{\text{Height(cm)} - 150}{4} \quad (2)$$

$$\text{Ideal Weight for women} = \text{Height(cm)} - 100 - \frac{\text{Height(cm)} - 150}{2.5} \quad (3)$$

$$\text{PNI} = 10 \times \text{Albumin} \left(\frac{\text{g}}{\text{dL}} \right) + 0.005 \times \text{Total Lymphocyte Count (per mm}^3) \quad (4)$$

Patient demographic and other clinical characteristics included in the analysis were age, gender, ethnicity, marital status, and vital signs such as heart rate, systolic blood pressure (SBP) and diastolic blood pressure (DBP). Additionally, comorbidities present upon ICU admission were collected, including hypertension, solid malignant tumor, diabetes, chronic heart failure, myocardial infarction, COPD, chronic neurologic disease, and cirrhosis. The Charlson comorbidity index (CCI) [34]

was utilized to assess the overall burden of comorbidities. The degree of organ dysfunction was quantified using the Sequential Organ Failure Assessment (SOFA) [35] score. We also included additional indices assessing the severity of illness at the time of ICU admission, such as Acute Physiology and Chronic Health Evaluation III (APSIII) [36], Organ Dysfunction and/or Infection (OASIS) [37], and Simplified Acute Physiology Score II (SAPS II) [38]. The first-day urine output in the ICU and laboratory test results at ICU admission, including hemoglobin levels, white blood cell (WBC) count, platelets count, blood urea nitrogen (BUN), albumin, lymphocyte count, and baseline SCr levels were included. Furthermore, we assessed the administration of nephrotoxic antimicrobials (i.e., glycopeptides, aminoglycosides, amphotericin B, and colistin), blood transfusion, and corticosteroid during the first day of ICU admission. Other relevant variables included the use of continuous renal replacement therapy (CRRT), mechanical ventilation, acquisition of multidrug-resistant organism (MDRO) infections, and the occurrence of septic shock during ICU stay.

Statistical analysis

We used a directed acyclic graph (DAG; Fig 2) to guide the selection of covariates for the multivariable models. The DAG was constructed based on theoretical considerations and clinical knowledge, aiming to identify confounders while avoiding overadjustment or adjusting for mediators. Covariates were selected to account for potential confounding effects, such as demographic variables, vital signs, SOFA score, and CCI score. We excluded variables such as nephrotoxic antimicrobial use, corticosteroid use, blood transfusions, and renal function from models, as these were considered potential mediators triggered by the clinical severity, which was already captured by variables such as laboratory test results, severity conditions, and nutritional scores. This approach supported clinically informed covariate selection without introducing bias from mediator adjustment.

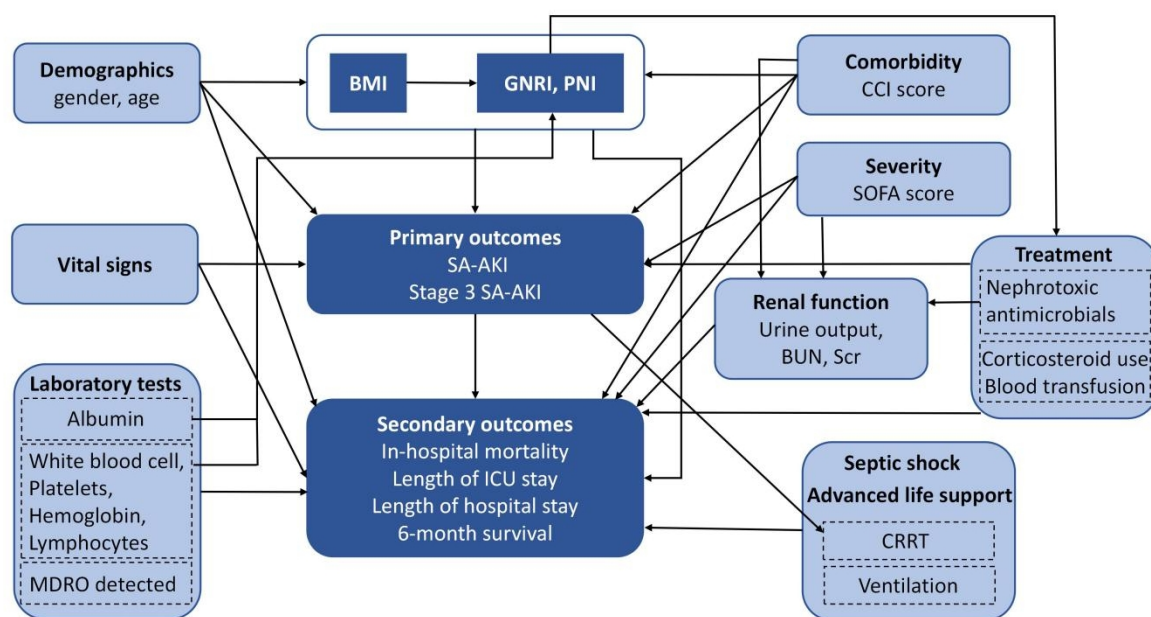


Fig 2 Directed Acyclic Graph

Abbreviations: BMI, body mass index; BUN, Blood urea nitrogen; CCI, Charlson comorbidity index; CRRT, continuous renal replacement therapy; GNRI, geriatric nutritional risk index; ICU, intensive care unit; MDRO, multidrug-resistant organism; PNI, prognostic nutritional index; SA-AKI, sepsis associated-acute kidney injury; Scr, Serum creatinine; SOFA, Sequential organ failure assessment.

Baseline characteristics were presented with continuous variables summarized using means and standard deviations (SDs) or medians and interquartile ranges (IQRs), while categorical variables expressed as counts and percentages. The prevalence of early SA-AKI, stage 3 SA-AKI, in-hospital mortality (among early SA-AKI patients), and 6-month survival after ICU admission (among early SA-AKI patients) were compared across BMI, GNRI, and PNI groups using the *chi-square* test. Additionally, the average lengths of ICU and hospital stays for SA-AKI patients were compared across these same groups using the *Kruskal-Wallis* test.

To investigate the association between BMI, GNRI, PNI, and outcomes, including early SA-AKI and stage 3 SA-AKI among older septic patients, we conducted a multivariable logistic regression analysis. The model was adjusted for significant demographic and clinical factors related to obesity, nutritional status, and SA-AKI, including gender, age, CCI score, SOFA score, vital signs, and laboratory test results (i.e., albumin level, WBC count, platelet count, hemoglobin, and lymphocyte count). To avoid collinearity and overadjustment, component variables of composite indices were not simultaneously included in the same model (e.g., albumin was excluded in

GNRI models, and both albumin and lymphocyte count were excluded in PNI models). To explore potential effect modification, we conducted interaction analyses by adding multiplicative interaction terms between each exposure (BMI categories, GNRI risk groups, and PNI) and clinically relevant modifiers (age group, sex, baseline SOFA category, and CCI category) in the adjusted models for early SA-AKI. Acknowledging the potential non-linear relationships between the independent variables and the outcomes of SA-AKI and in-hospital mortality among early SA-AKI patients, multivariable fractional polynomial models and restricted cubic spline curves were then introduced to better capture these complexities. The empirically derived values of BMI/GNRI/PNI at which the fitted curve crossed the null (i.e., where the estimated log-odds was closest to zero, corresponding to an *odds ratio* (*OR*) of ≈ 1) were also calculated.

For mortality outcomes, in-hospital mortality was analyzed using logistic regression, while survival analysis was performed for 6-month mortality. To assess survival probabilities, Kaplan-Meier estimates and log-rank tests were used to compare the cumulative mortality and survival rates across BMI, GNRI, PNI groups. The association between BMI, GNRI, PNI and 6-month survival in early SA-AKI patients was examined using Cox regression, adjusting for gender, age, CCI score, SOFA score, vital signs, laboratory test results (i.e., albumin level, WBC count, platelet count, hemoglobin, lymphocyte count, and MDRO detection), septic shock, and mechanical ventilation. Additionally, a multivariable linear regression model was employed to explore the relationship between BMI, GNRI, PNI, and the length of ICU and hospital stays in early SA-AKI patients, with the same covariates included in the models. The impact of the independent variables was quantified using effect sizes derived from various regression models, including *Odds Ratios* (*OR*), *Hazard Ratios* (*HR*), and regression coefficients (β), each accompanied by their respective 95% *Confidence Intervals* (*CI*s).

The missing data proportions for most of the key variables were between 0 and 0.6%, except for albumin (57.8%) and lymphocyte count (34.1%), which suggests minimal potential for bias due to missing data. Due to missing data in albumin and lymphocyte count, analyses involving GNRI and PNI were conducted as sub-cohort analyses. The Little's missing completely at random (MCAR) test was performed to examine the missing data pattern ($P < 0.001$), suggesting that the missing data is not MCAR. Logistic regression for missingness indicators indicated that the missing

albumin and lymphocyte data were associated with variables such as septic shock, disease severity, and comorbidity burden, suggesting that the missing data is likely missing at random. Multiple imputation was therefore performed to handle the missing data. We applied predictive mean matching for multiple imputation using the *mice* function in R. The threshold for statistical significance was set at a *P*-value of < 0.05 , using a two-tailed test. All statistical analyses were performed using R software (Version 4.3.3).

Results

Patient characteristics

A total of 4238 older septic patients without pre-existing renal dysfunction were included in the analysis (Fig 1). As shown in Table 1, the average age was approximately 77 years, with over half of them being older than 75. More than 70% of the patients identified as white. The median CCI score was 6 (IQR: 4, 7), and the most prevalent comorbidities were hypertension (58.8%), solid malignant tumors (27.9%), diabetes (24.6%), and congestive heart failure (21.5%). The median SOFA score was 6 (IQR: 4, 8). The average BMI of older patients with sepsis was 27.6, with approximately 68.4% classified as overweight, obese, or severely obese. The median GNRI score of 98.6, indicating no nutritional risk on average, though over 30% of patients had moderate to major nutritional risk. The median PNI score was 36.5 (IQR: 31.0, 42.5). During ICU stay, approximately 18.6% of the patients developed septic shock, 4.7% acquired an MDRO infection, around 70% required mechanical ventilation, and 6% required CRRT.

Table 1 Baseline characteristics of the older septic patients without pre-existing renal dysfunction (N= 4238)

Characteristics	N (%) / Median (IQR)	Characteristics	N (%) / Median (IQR)
Age (Mean $\pm$ SD)	76.95 $\pm$ 7.63	OASIS score	36 (31, 42)
65~74	1881 (44.4)	SAPS II score	43 (36, 52)
75~84	1619 (38.2)	BMI	27.6 (24.0, 32.0)
85~	738 (17.4)	Underweight	135 (3.2)
Sex (Female)	1915 (45.2)	Normal weight	1201 (28.3)
Race		Overweight	1426 (33.6)

White	3010 (71.0)	Obesity	1186 (28.0)
African	235 (5.6)	Severe obesity	290 (6.8)
Asian	111 (2.6)	GNRI (n=1786) ^a	98.6 (88.8, 110.0)
Hispanic/Latino	86 (2.0)	No risk	926 (51.8)
Other	132 (3.1)	Low risk	271 (15.2)
Unknown	664 (15.7)	Moderate risk	352 (19.7)
Marriage		Major risk	237 (13.3)
Married	2085 (49.2)	PNI (n=1423)	36.5 (31.0, 42.5)
Widowed	768 (18.1)	Laboratory tests	
Single	649 (15.3)	WBC (K/uL; n=4236)	12.5 (9.4, 16.2)
Divorced	301 (7.1)	Hemoglobin (g/dL; n=4236)	10.4 (9.3, 11.8)
Vital signs		Platelets (K/uL; n=4236)	176.5 (130.5, 240.1)
Heart rate	83.1 (75.2, 93.8)	BUN (mg/dL; n=4234)	21.0 (15.5, 32.0)
SBP (n=4237)	112.2 (105.2, 122.2)	Albumin (g/dL; n=1786)	3.1 (2.6, 3.6)
DBP (n=4237)	57.5 (52.4, 63.5)	Lymphocytes (K/uL; n=2794)	1.1 (0.7, 1.7)
Comorbidities		SCr (mg/dL; n=4235)	0.7 (0.6, 0.9)
Hypertension	2490 (58.8)	MDRO detected ^b	201 (4.7)
Solid malignant tumors	1182 (27.9)	UO first-day (mL; n=4213)	1437.0 (918.0, 2115.0)
Diabetes	1043 (24.6)	Septic shock ^b	787 (18.6)
Congestive heart failure	912 (21.5)	Treatment ^b	
Myocardial infarction	718 (16.9)	Corticosteroid use	487 (11.5)
COPD	292 (6.9)	Blood Transfusion	1194 (28.2)
Chronic neurologic	264	Nephrotoxic	2048 (48.3)

disease	(6.2)	antimicrobials	
Cirrhosis	190 (4.5)	Advanced life support	
		^b	
CCI score	6 (4, 7)	CRRT	256 (6.0)
SOFA score	6 (4, 8)	Ventilation	3110 (73.4)
APSIII score	47 (35, 63)		

Abbreviations: APSIII, Acute Physiology and Chronic Health Evaluation III; BMI, Body mass index; BUN, Blood urea nitrogen; CCI, Charlson comorbidity index; COPD, Chronic obstructive pulmonary disease; CRRT, Continuous renal replacement therapy; DBP, Diastolic blood pressure; GNRI, Geriatric nutritional risk index; IQR, Interquartile range; MDRO, Multidrug-resistant organism; OASIS, Organ Dysfunction and/or Infection; PNI, Prognostic nutritional index; SAPS II, Simplified Acute Physiology Score II; SBP, systolic blood pressure; SCr, Serum creatinine; SD, Standard deviation; SOFA, Sequential Organ Failure Assessment; UO, Urine output; WBC, White blood cell.

Notes: ^a The score range of different GNRI groups are: No risk: 98 to 212; Low risk: 92 to 98; Moderate risk: 82 to 92; Major risk: 55 to 82. ^b Performed/acquired during hospitalization.

Obesity, nutritional status, and early SA-AKI in older septic patients without renal dysfunction

Among the cohort (n=4238), 46.4% (n=1967) developed SA-AKI within 48 hours of ICU admission, with 11.1% (n=471) of cases classified as stage 3 SA-AKI. The incidence of both SA-AKI and stage 3 SA-AKI was significantly higher in obese (SA-AKI: adjusted odds ratio [AOR]=1.95, 95% CI: 1.63 to 2.35; stage 3 SA-AKI: AOR=1.43, 95% CI: 1.06 to 1.93) and severely obese patients (SA-AKI: AOR=2.19, 95% CI: 1.63 to 2.94; stage 3 SA-AKI: AOR=1.72, 95% CI: 1.14 to 2.60) compared to normal-weight patients (Table 2). Overweight patients also had a significantly higher risk of early SA-AKI (AOR=1.39, 95% CI: 1.17 to 1.65). The multivariable fractional polynomial regression models treating BMI as a continuous variable yielded results consistent in direction and significance with the categorical BMI analyses, showing a 4% increase in the adjusted risk of early SA-AKI for every 1-point increase in BMI (95% CI: 1.03 to 1.05; Table S1-S2, Fig S1-S2).

In the sub-cohort of patients with available GNRI data, univariable analyses of GNRI and its association with SA-AKI and stage 3 SA-AKI yielded inconsistent results. The *chi-square* test revealed no significant correlation, while non-linear models showed a negative correlation with SA-AKI and a positive correlation with stage 3 SA-AKI. However, after multivariable adjustment, no statistically significant associations were found between GNRI and the likelihood of SA-AKI or stage 3 SA-

AKI (Table 2, Table S1-S2, Fig S1-S2). Additionally, although higher BMI was associated with a higher GNRI score across BMI groups (Table S5), no significant interactions between GNRI and BMI were found (Table S6). However, several statistically significant but counterintuitive interactions emerged, including GNRI (low risk group) $\times$ age ($AOR=0.95$, 95% CI : 0.91 to 0.99), GNRI (major risk group) $\times$ gender male ($AOR=0.36$, 95% CI : 0.18 to 0.71), and GNRI (moderate risk group) $\times$ CCI ($AOR=0.83$, 95% CI : 0.74 to 0.95) (Table S6). Similarly, in the sub-cohort of patients with available PNI data, although higher PNI was associated with a lower risk of SA-AKI and stage 3 SA-AKI in univariable analyses, these associations were no longer statistically significant after multivariable adjustment (Table 2; Table S1-S2; Fig S1-S2).

Table 2 Associations of obesity, nutritional status (sub-cohort analyses), and early SA-AKI in older septic patients without renal dysfunction

	Stage 1, 2, or 3 SA-AKI (n=1967)		Stage 3 SA-AKI (n=471)	
	Participants [number (%)]	Adjusted OR (95% CI)	Participants [number (%)]	Adjusted OR (95% CI)
BMI ^a				
Under weight	47 (34.8)	0.68 (0.45, 1.03)	14 (10.4)	0.93 (0.49, 1.77)
Normal weight	465 (38.7)	Reference	97 (8.1)	Reference
Overweight	662 (46.4)	1.39 (1.17, 1.65) ***	156 (10.9)	1.31 (0.98, 1.75)
Obesity	631 (53.2)	1.95 (1.63, 2.35) ***	148 (12.5)	1.43 (1.06, 1.93) *
Severe obesity	162 (55.9)	2.19 (1.63, 2.94) ***	56 (19.3)	1.72 (1.14, 2.60) *
P value ^b	<0.001	----	<0.001	----
GNRI ^c				
Major risk	112 (47.3)	0.68 (0.46, 1.02)	45 (19.0)	1.02 (0.59,

				1.75)
Moderate risk	182 (51.7)	0.89 (0.64, 1.23)	58 (16.5)	0.94 (0.60, 1.46)
Low risk	145 (53.5)	1.03 (0.75, 1.43)	47 (17.3)	1.14 (0.73, 1.79)
No risk	496 (53.6)	Reference	149 (16.1)	Reference
<i>P</i> value ^b	0.360	----	0.747	----
PNI ^d				
First quartile	213 (59.8)	Reference	92 (25.8)	Reference
Second quartile	206 (57.9)	1.17 (0.83, 1.64)	73 (20.5)	1.04 (0.69, 1.56)
Third quartile	172 (48.5)	1.06 (0.75, 1.50)	50 (14.1)	0.99 (0.63, 1.55)
Fourth quartile	166 (46.6)	1.11 (0.77, 1.58)	43 (12.1)	0.91 (0.56, 1.47)
<i>P</i> value ^b	<0.001	----	<0.001	----

Abbreviations: BMI, Body mass index; GNRI, Geriatric nutritional risk index; OR, Odds ratios; PNI, Prognostic nutritional index; SA-AKI, Sepsis-associated acute kidney injury.

Note: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$; ^a Adjusted for gender, age, CCI, SBP, SOFA, albumin level, white blood cell count, platelet count, hemoglobin, and lymphocyte count. ^b *P*value of *Chi-square* test.

^c Adjusted for gender, age, BMI, CCI, SBP, SOFA, white blood cell count, platelet count, and hemoglobin, and lymphocyte count with the sub-cohort of patients with available GNRI data. ^d Adjusted for gender, age, BMI, CCI, SBP, SOFA, white blood cell count, platelet count, and hemoglobin with the sub-cohort of patients with available PNI data.

Obesity, nutritional status, and outcomes in older patients with early SA-AKI

Among older patients with early SA-AKI (n=1967; Table S3), 24.5% (n=482) died during hospitalization, and 52.3% (n=1029) died within 6 months post ICU admission. In-hospital mortality varied across BMI categories, with the highest mortality observed in the underweight patients (46.8%), compared to 22.0%–27.3% in other BMI categories (Table 3). Restricted cubic spline analysis revealed a statistically significant nonlinear association between BMI and in-hospital mortality (P for nonlinearity = 0.009), with risk increasing at very low BMI values (BMI < 14.15; Fig 3). After multivariable adjustment, underweight patients remained at significantly higher risk of in-hospital mortality compared to normal-weight patients

(*AOR* = 2.06, 95% *CI*: 1.07 to 3.96) (Table 3).

For 6-month mortality among patients with early SA-AKI, underweight remained independently associated with a higher risk of mortality (Adjusted hazard ratios [*AHR*] = 1.88, 95% *CI*: 1.21 to 2.92), while severely obese patients had a significantly reduced risk compared with normal-weight patients (*AHR* = 0.67, 95% *CI*: 0.48 to 0.95; Table 3; Fig 4). Although overweight/obesity groups showed lower 6-month mortality in univariate comparisons, these differences were not statistically significant in the adjusted Cox model. Regarding resource use, the median ICU and hospital length of stay among patients with early SA-AKI was 5 and 11 days, respectively; length of stay differed by BMI category, with the longest stays observed in the obesity/severe obesity group (about 7 days in ICU and 12 days in the hospital; Table S4).

Table 3 Outcomes of patients with early SA-AKI (n=1967) across different BMI and nutritional status

	In-hospital mortality (n=482)		6-month mortality after ICU admission (n=1029)		
	n (%)	Adjusted <i>OR</i> (95% <i>CI</i>)	n (%)	Log-rank test	Adjusted <i>HR</i> (95% <i>CI</i>)
	^b		^b		
BMI					
Under weight	22 (46.8)	2.06 (1.07, 3.96)*	43 (91.5)	0.005	1.88 (1.21, 2.92)**
Normal weight	127 (27.3)	Reference ^c	279 (60.0)	Reference	Reference ^c
Overweight	152 (23.0)	0.77 (0.57, 1.03)	336 (50.8)	0.215	0.87 (0.70, 1.08)
Obesity	139 (22.0)	0.71 (0.52, 0.96)*	294 (46.6)	0.274	0.88 (0.70, 1.11)
Severe obesity	42 (25.9)	0.65 (0.41, 1.02)	77 (47.5)	0.024	0.67 (0.48, 0.95)*
<i>P</i> value ^a	0.001	----	<0.001	<0.001	----
GNRI					
Major risk	50 (44.6)	1.80 (1.00, 3.22)*	85 (75.9)	0.018	1.38 (1.06, 1.81)*

Moderate risk	56 (30.8)	1.09 (0.69, 1.74)	119 (65.4)	0.551	1.06 (0.87, 1.31)
Low risk	45 (31.0)	1.28 (0.81, 2.03)	83 (57.2)	0.949	0.99 (0.81, 1.22)
No risk	139 (29.3)	Reference ^d	268 (56.5)	Reference	Reference ^d
<i>P</i> value ^a	0.01	----	<0.00	<0.00	----
	9		1	1	
PNI					
First quartile	86 (40.6)	Reference ^e	145 (68.4)	Reference	Reference ^e
Second quartile	59 (28.8)	0.61 (0.39, 0.93) *	123 (60.0)	0.088	0.84 (0.69, 1.03)
Third quartile	54 (32.0)	0.76 (0.48, 1.20)	101 (59.8)	0.145	0.85 (0.69, 1.06)
Fourth quartile	43 (27.2)	0.71 (0.43, 1.17)	83 (52.5)	0.072	0.81 (0.65, 1.02)
<i>P</i> value ^a	0.022	----	0.020	<0.00	----
				1	

Abbreviations: HR, hazard ratio; BMI, Body mass index; GNRI, Geriatric nutritional risk index; HR, Hazard ratios; ICU, Intensive care unit; OR, Odds ratios; PNI, Prognostic nutritional index; SA-AKI, Sepsis-associated acute kidney injury.

Note: ^a *P* value for Chi-square tests. ^b Number of mortalities varied due to missing data on the GNRI and PNI. ^c Adjusted for age, gender, CCI, SOFA, SBP, albumin level, hemoglobin, white blood cell count, platelets count, lymphocyte count, mechanical ventilation, acquisition of MDRO infection during hospitalization, septic shock during hospitalization. ^d Adjusted for age, gender, CCI, SOFA, SBP, hemoglobin, white blood cell count, platelets count, lymphocyte count, mechanical ventilation, acquisition of MDRO infection during hospitalization, septic shock during hospitalization. ^e Adjusted for age, gender, CCI, SOFA, SBP, hemoglobin, white blood cell count, platelets count, mechanical ventilation, acquisition of MDRO infection during hospitalization, septic shock during hospitalization. **P*<0.05, ***P*<0.01.

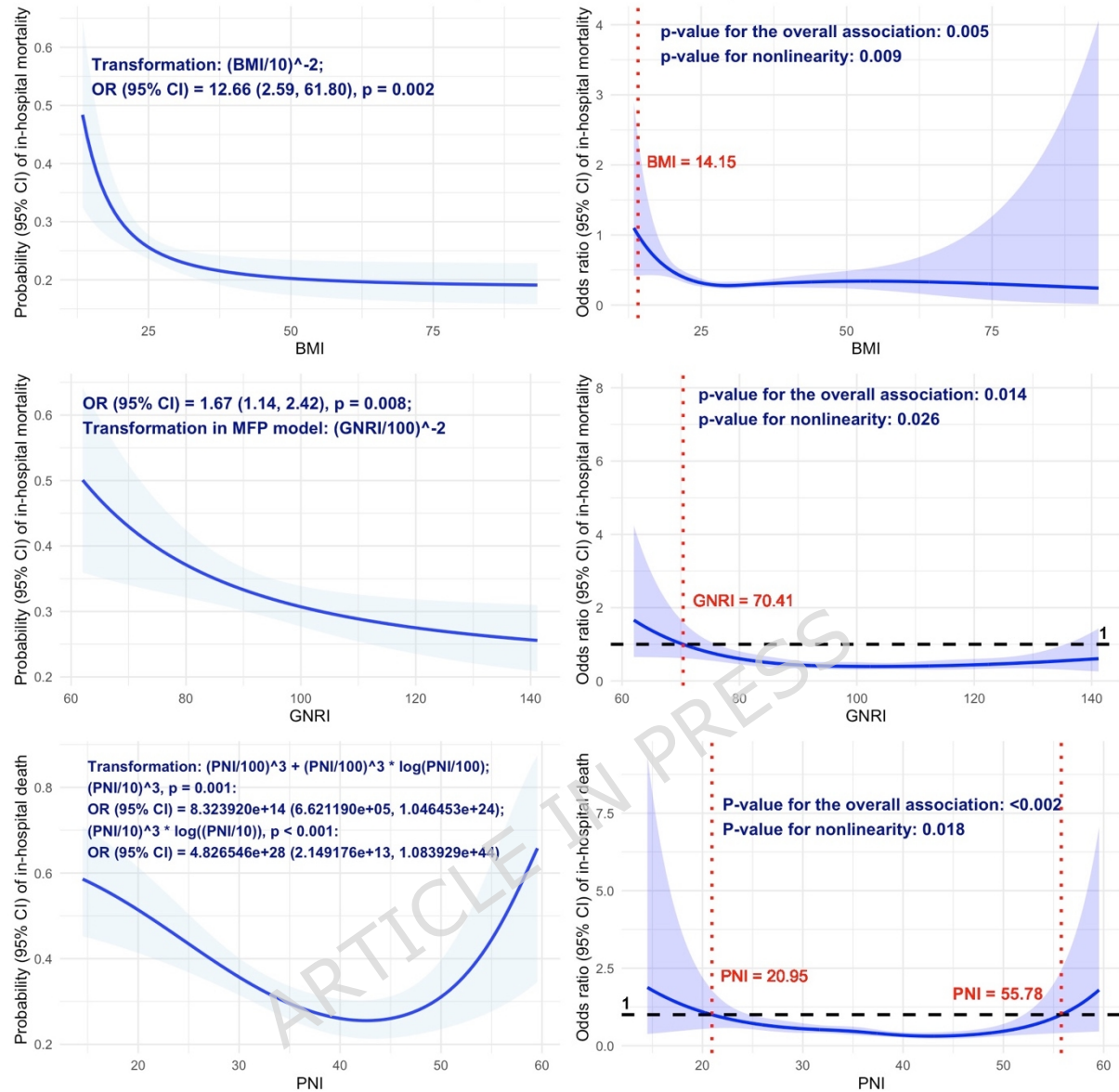


Fig 3 Estimated probability (left) and odds ratios (right) of in-hospital mortality across BMI, GNRI, and PNI

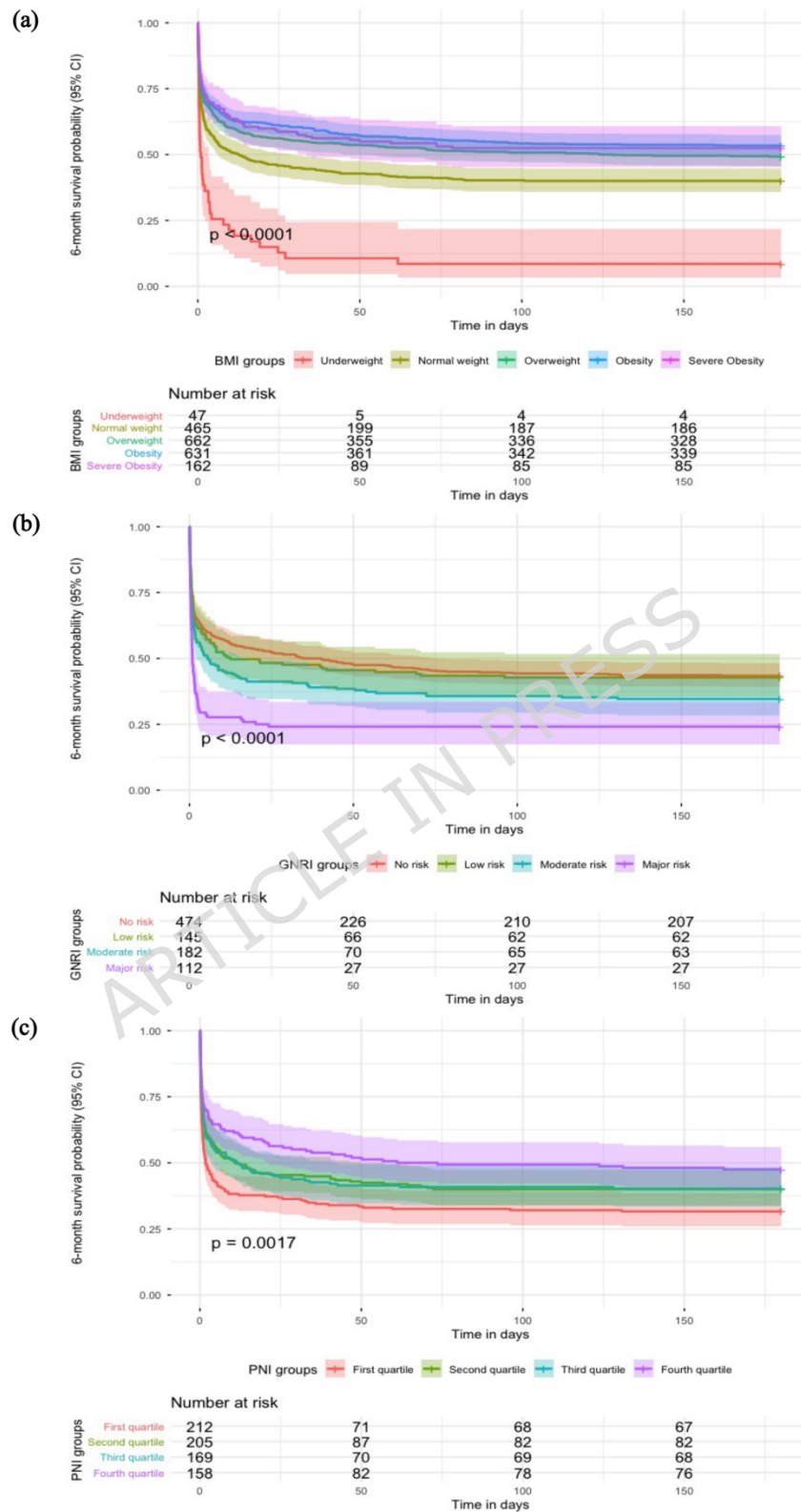


Fig 4 Survival curves of older patients with early sepsis-associated acute kidney

injury across BMI (a), GNRI (b), and PNI (c) groups.

For GNRI, a nonlinear relationship with in-hospital mortality was found (P -value for nonlinearity=0.026), with mortality risk increasing below a GNRI score of 70.41 (Fig 3). The major GNRI risk group had worse 6-month survival ($AHR=1.38$, 95% CI : 1.06 to 1.81) compared to the no risk group (Table 3). There were no significant differences in ICU or hospital stay durations across GNRI groups (Table S4).

Higher PNI quartiles were associated with reduced risk of in-hospital and 6-month mortality in univariate analyses (Table 3; Fig 4). The restricted cubic spline analysis revealed a nonlinear association (P -value for nonlinearity=0.018), suggested an increased risk of in-hospital mortality at PNI scores below 20.95 and above 55.78 (Fig 3). After adjustment, the second PNI quartile (vs. the first) had a reduced risk of in-hospital mortality ($AOR=0.61$, 95% CI : 0.39 to 0.93). Furthermore, no significant differences in ICU or hospital stay durations were observed across PNI groups (Table S4).

Discussion

This study aimed to investigate the associations between BMI, GNRI, and PNI with the development of early SA-AKI and subsequent mortality in older septic ICU patients without pre-existing renal dysfunction. Our findings suggest that older septic patients who are overweight, obese and severely obese are associated with a significantly higher risk of developing early SA-AKI. However, once SA-AKI occurs, underweight individuals are related to a higher risk of both in-hospital and 6-month mortality. Although no statistically significant relationship was found between GNRI and the incidence of early SA-AKI after adjustment for confounding variables, a major-risk GNRI score was associated with an increased risk of in-hospital mortality and 6-month mortality. While no direct association between PNI and early SA-AKI was found, we identified a concave relationship (P -value for nonlinearity = 0.018), where both low and high PNI scores were related to an increased risk of in-hospital mortality. Within this selected cohort of older septic ICU patients without documented CKD stage 3-5 or elevated SCr at ICU admission, these findings provide valuable insights into the early identification of high-risk older patients with early SA-AKI.

Our study aligns with previous research that have shown obesity is associated with a higher risk of early SA-AKI [25, 39]. A multicenter study also found that higher BMI is independently associated with increased AKI severity [40]. Recent evidence

suggests that microvascular dysfunction, inflammation, and metabolic reprogramming play key roles in the development of SA-AKI [4]. Obesity induces structural, hemodynamic, and metabolic changes in the kidney, activates the renin-angiotensin-aldosterone system, and may contribute to proteinuric renal injury [41, 42]. Additionally, oxidative stress may partially mediate the link between obesity and AKI [43]. In acute conditions like SA-AKI, obesity fails to mitigate the immediate physiological impacts, such as inflammation and organ dysfunction, which can ultimately worsen outcomes [44, 45].

When examining prognosis after early SA-AKI in older patients, underweight individuals were found to have a significantly higher risk of both in-hospital and 6-month mortality, aligning with previous studies in septic or chronically critically ill patients [46–48]. Overweight and obese patients initially showed lower unadjusted mortality, but some of these protective associations (e.g., overweight and mortality) did not remain statistically significant after multivariable adjustment, as also observed in other studies [49, 50]. Thus, our results suggest that the apparent “obesity paradox”, which suggests obesity may offer protective benefits in critical illness, is mainly observed in unadjusted survival comparisons and should be interpreted with caution. This discrepancy may be due to residual confounding (e.g., medication use), heterogeneity in obesity phenotypes (e.g., sarcopenic obesity in older adults), and limitations of BMI as a proxy for body composition (e.g., the absence of a universally accepted BMI threshold specifically validated for older adults leading to potential misclassification). Further research using more precise measures of body composition is needed to further validate these findings [51, 52].

Our results did not reveal a statistically significant relationship between GNRI and early SA-AKI incidence after multivariable adjustment, contrasting with previous studies in older postoperative populations that identified higher GNRI as a predictor of AKI [53, 54]. This discrepancy may be partly explained by the acute inflammatory response in sepsis, which can alter albumin levels and limit the reliability of GNRI as a nutritional marker in early sepsis. Additionally, fluid shifts due to oedema, fluid overload, or severe dehydration may distort GNRI values, potentially masking underlying nutritional conditions and attenuating its association with SA-AKI. Unmeasured factors such as frailty and sarcopenia may further contribute to residual confounding and complicate interpretation [55]. Longitudinal nutritional assessments for septic older individuals are expected to more accurately

characterize their chronic malnutrition and validate the findings. Exploratory interaction analyses yielded several statistically significant but clinically counterintuitive findings; however, these showed no consistent pattern and should not be overinterpreted. We therefore did not draw substantive conclusions regarding effect modification from these analyses. GNRI did show prognostic relevance in older patients with SA-AKI. Consistent with previous studies [21, 56], our study found that a lower GNRI is significantly associated with increased mortality risk, particularly at very low values (GNRI <70.41). Although this threshold was derived from data-driven analyses and should be interpreted cautiously, it likely reflects severely compromised nutritional reserves. Further prospective studies are needed to validate whether this threshold can serve as a reliable marker for identifying patients who may benefit from proactive nutritional strategies.

In our study, while higher PNI was significantly associated with a lower risk of early SA-AKI in univariate analysis, this association was no longer statistically significant after multivariable adjustment, indicating that PNI may not be an independent predictor of early SA-AKI in older septic patients without pre-existing renal dysfunction. However, PNI was significantly associated with mortality among older patients with early SA-AKI. The risk of in-hospital mortality increased not only in patients with severely impaired nutritional and immune function (PNI<20.95), but also in those with excessively high PNI (>55.78), potentially indicating an overactive immune response or excessive therapeutic intervention. This suggests that PNI may be more informative for mortality risk stratification than for predicting SA-AKI onset [57]. Since PNI is derived from serum albumin and lymphocyte count, it is influenced by inflammation and immune dysregulation in sepsis [58, 59]. While PNI reflects both nutritional and immune status, it may not fully capture the complex metabolic and inflammatory processes underlying sepsis, such as oxidative stress [60]. Therefore, PNI should be interpreted cautiously and ideally used alongside other biomarkers and clinical indicators when assessing prognosis in septic patients. The finding that GNRI and PNI are correlated with mortality but not with the initial onset of early SA-AKI, is not contradictory. Instead, it underscores the distinct pathophysiological roles that nutritional and immune reserves assume at various stages of critical illness. This discrepancy may reflect the fact that early SA-AKI is primarily driven by acute hemodynamic instability and inflammatory responses,

whereas GNRI and PNI capture baseline nutritional and immune reserve, which are more relevant to recovery and survival after organ injury.

This study contributes to the literature in three main ways. First, it focuses on older septic ICU patients without pre-existing renal dysfunction, a group characterized by reduced renal reserve and distinct body-composition phenotypes. Second, it integrates BMI with two complementary nutritional indices (GNRI and PNI), offering a multidimensional approach to characterizing adiposity and nutritional-immune vulnerability. Third, by distinguishing between early SA-AKI onset and post-AKI mortality, it highlights stage-specific risk patterns that may inform future risk-stratification strategies.

Several limitations should be noted. First, this study was based on a single public critical care database (MIMIC-IV) without external validation, which may limit generalizability; its retrospective design is inherently susceptible to residual confounding and does not permit causal inference. Second, due to limited availability of longitudinal pre-admission kidney function data, pre-existing CKD was mainly identified through documented diagnoses, which may have misclassified some patients with undiagnosed stage 3 CKD and potentially affected generalizability. Third, AKI was defined based solely on SCr due to the systemic lack of reliable hourly urine output data in MIMIC-IV, which may have led to underestimation of oliguric AKI. In addition, fluid resuscitation in sepsis can induce SCr dilution or post-resuscitation rebound, which may introduce misclassification of AKI onset. To reduce misclassification of pre-existing renal dysfunction as incident early SA-AKI, we excluded patients with documented CKD stage 3–5 and those with elevated SCr at ICU admission. However, SCr is influenced not only by kidney function but also by SCr generation, which depends on muscle mass, nutritional status, and frailty. This issue is especially relevant in older adults. Therefore, conditioning study inclusion on baseline SCr may have introduced structural selection bias and preferentially retained patients with low muscle mass, sarcopenia, or poor nutritional reserve, whose SCr may remain low despite reduced renal reserve, while excluding patients with relatively preserved muscle mass and higher creatinine generation. Because this selection mechanism is related to both the exposures of interest and SA-AKI classification, it may have affected the internal validity of the estimated associations and limited their generalizability to the broader population of older septic patients, and the observed associations should be

interpreted within this selected population. Future studies with comprehensive urine output and baseline renal function data are warranted. Fourth, generalizability is also limited by cohort characteristics, including a high prevalence of overweight/obesity and the use of U.S. BMI cutoffs, which may not apply to other populations (e.g., Asian populations). Fifth, several important confounders (e.g., fluid balance, vasopressor exposure, primary sources of sepsis, and timing/adequacy of antibiotics) were unavailable, potentially introducing residual confounding—particularly in the associations between nutritional indices (GNRI, PNI) and early SA-AKI risk. Finally, GNRI and PNI were calculated using albumin, lymphocyte count, and weight measured at ICU admission during active sepsis, which likely reflect not only pre-existing nutritional reserve but also acute inflammatory responses and fluid shifts. Therefore, our findings should be interpreted as prognostic associations between nutrition-inflammation status at ICU admission and early SA-AKI, rather than evidence of a causal effect of chronic malnutrition on AKI risk. The absence of pre-sepsis or longitudinal nutritional assessments, such as pre-albumin and body composition metrics, is an important limitation that should be addressed in future prospective studies involving critically ill older patients.

Conclusions

In a selected cohort of older septic ICU patients without pre-existing renal dysfunction, overweight and obesity were independently associated with a higher risk of early SA-AKI. Among those who developed early SA-AKI, underweight and poorer nutritional status—reflected by lower GNRI and PNI—were associated with worse survival up to 6 months. These findings underscore the prognostic value of BMI and nutritional indices in identifying patients at differential risk for early SA-AKI onset and subsequent mortality in this selected population. Further prospective studies are warranted to validate these associations and explore their potential implications for risk stratification.

List of abbreviations

AHR: Adjusted hazard ratios

AKI: Acute kidney injury

AOR: Adjusted odds ratios

APSIII: Acute Physiology and Chronic Health Evaluation III

BMI: Body mass index

BUN: Blood urea nitrogen

CCI: Charlson comorbidity index
 CI: Confidence interval
 CKD: Chronic kidney disease
 COPD: Chronic obstructive pulmonary disease
 CRP: C-reactive protein
 CRRT: Continuous renal replacement therapy
 DAG: Directed acyclic graph
 DBP: Diastolic blood pressure
 GNRI: Geriatric nutritional risk index
 HR: Hazard ratios
 ICU: Intensive care unit
 IQR: Interquartile range
 KDIGO: Kidney Disease Improving Global Outcomes
 MCAR: Missing completely at random
 MDRO: Multidrug-resistant organism
 MIMIC IV: Medical Information Mart for Intensive Care IV
 OASIS: Organ Dysfunction and/or Infection
 OR: Odds ratios
 PNI: Prognostic nutritional index
 RRT: Renal replacement therapy
 SA-AKI: Sepsis-associated acute kidney injury
 SAPS II: Simplified Acute Physiology Score II
 SBP: Systolic blood pressure
 Scr: Serum creatinine
 SD: Standard deviation
 SOFA: Sequential Organ Failure Assessment
 UO: Urine output
 U.S.: The United States
 WBC: White blood cell
 WHO: World Health Organization

Statements and Declarations

Ethics approval and consent to participate

The MIMIC-IV databases are publicly available, and researchers who agree to the data use agreement and have completed "protecting human subjects training" can

request access. The MIMIC database was approved by the institutional review boards of the Beth Israel Deaconess Medical Center (2001-P-001699/14) and the Massachusetts Institute of Technology (No.0403000206), which waived the requirement for individual patient consent because the datasets contained deidentified information. This study was conducted in accordance with the Declaration of Helsinki.

Consent for publication

Not applicable.

Availability of data and materials

This research was conducted using the MIMIC-IV databases. This data can be found at: MIMIC-IV v2.2 (<https://physionet.org/content/mimiciv/2.2/>)

Competing Interests

The authors declare that they have no competing interests.

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Authors' contributions

DZS, KZH, and YSL equally contributed to the data interpretation and the drafting of the original manuscript. DZS and YJY contributed to data acquisition. XL contributed to the data analysis, data interpretation, figures manipulation, and acquired funding. XL, SFL, and YJY critically reviewed and revised the manuscript, and were considered co-corresponding authors. All authors contributed to the study concept and design. All authors read and approved the final manuscript for publication.

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